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Paleomagnetism of the Late Cretaceous volcanic rocks of the Shimaoshan Group in Yongtai County, Fujian Province

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It has been debated whether there was southward movement of the South China Block (SCB) during the Cretaceous. In this study, a paleomagnetic investigation was carried out on the Late Cretaceous volcanic rocks (~88 Ma) of the Shimaoshan Group in Yongtai County, Fujian Province. Rock magnetic experiments showed that magnetite in pseudo-single-domain and multi-domain grain and hematite were predominant magnetic phases. Stepwise thermal demagnetization successfully isolated characteristic directional components at high-temperature interval (> 500°C) from 383 specimens in 19 sites, which yielded a paleomagnetic pole for the studied section at 83.1°N, 152.6°E (N = 19, A_{95} = 3.9°), and the scatter S_B = 9.0. The Fisher distribution of virtual geomagnetic poles (VGPs) and the consistence of S_B with the expected value at the 95% confidence level indicate that the yielded paleomagnetic pole is free of paleomagnetic secular variation influence. The new pole, which is well consistent with that from the Eurasian apparent polar wander path (APWP) curve, suggests no obvious southward movement of the sampling site during the Cretaceous.

South China Block, Late Cretaceous, volcanic rock, paleomagnetic secular variation, paleomagnetic pole

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The South China Block (SCB), situated at the southeastern margin of the Eurasian plate, plays an important role in China continent's tectonic evolution. Paleomagnetic studies have suggested that SCB and NCB (North China Block) began to amalgamate in the Late Permian, forming the Qinling-Dabie orogenic belt, and converged together as a rigid plate no later than the Middle to Late Jurassic [1–5]. By the closure of the paleo-Asian Ocean and Mongol-Okhotsk Ocean, these Chinese blocks had been assembled with Eurasian plate in the Early Cretaceous [6–9]. However, it remains debatable whether SCB underwent southward movement during the Cretaceous [10–16]. For example, some paleomagnetic results of the red beds showed no obvious latitudinal movement relative to Eurasian plate during this period [13, 14], whereas other paleomagnetic data indicated obvious southward movement in the Late Cretaceous [10–12, 15, 16]. Laboratory redeposition and compaction experiment revealed that depositional or post-depositional detrital remanent magnetization (DRM or pDRM) under some circumstances might suffer from inclination flattening, causing the underestimate of the block's paleolatitude [17–20]. It has been found that the inclination from Central Asia's Tertiary red beds is ~20° lower than that of the expected [21]. To exclude the inclination shallow effects of the sedimentary rocks, paleomagnetic data of volcanic rocks are essential to constrain the block's paleogeographic position.

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Intense magmatic activity occurred during the Mesozoic in the eastern part of SCB and produced a large number of medium to acid volcanic rocks [22–25]. Bimodal volcanism also existed in the Shimaoshan Group in some areas, which produced basic basalt and acidic rhyolite in interbedded sequences [26]. Wang et al. [27] used the high-resolution $^{40}Ar/^{39}Ar$ dating and determined the precise age of the Shimaoshan Group in Fujian Province (87.9–88.6 Ma) and the Moshishan Group in eastern Zhejiang Province (109–118 Ma), indicative of formation in the Cretaceous Normal Superchron (CNS). In this study we carried out a paleomagnetic and rock magnetic investigation on a consecutive volcanic sequence of the Shimaoshan Group in Yongtai County, Fujian Province, to determine the SCB's paleolatitude and discuss the paleomagnetic secular variation (PSV) within CNS.

1 Paleomagnetic sampling

The Yongtai section (25.7°N, 119°E) is a type section for the Shimaoshan Group, which overlies on volcanic rocks of the Nanyuan Formation and the Fankeng Formation. The bimodal volcanic rocks consist mainly of gray-green and gray-purple rhyolite, basaltic andesite, and dense massive basalt (Figure 1). $^{40}Ar/^{39}Ar$ ages from the top and the middle of the section were 87.9 ± 0.6 and 88.1 ± 0.8 Ma [27], respectively.

Paleomagnetic samples were drilled using a portable gasoline-power diamond drill along a road from Yongtai to Putian County, Fujian Province. In total, 418 samples (23 sites) were collected. Cores were orientated in situ using a magnetic compass or simultaneously using a sun compass if possible. The result of the sun compass was indistinguishable from that measured by the magnetic compass and then corrected by the International Geomagnetic Reference Field, suggesting no local magnetic anomalies. The lithologies of sampling sites were dense massive basalt (YT1-YT4B), crystal ignimbrite (YT7-YT9), basaltic andesite (YT13A-YT18), and gray-green rhyolite (YT20A-YT24) from bottom to top. Samples were cut into 2.2-cm-high and 2.54 cm diameter cylinder in lab for paleomagnetic measurement.

2 Experimental results

2.1 Rock magnetic investigation

Rock magnetic measurements were conducted on 34 representative samples to examine the type of magnetic minerals. The magnetic hysteresis loop, stepwise isothermal remanent magnetization (IRM) acquisition up to a field of ~1 T and back-field demagnetization of IRM at room temperature were conducted on a Variable Field Translation Balance System (VFTB). Temperature-dependence susceptibility (κ-T) experiments were performed in an argon atmosphere using a KLY-3s (CS-3) with heating and cooling rate of ~13°C min^{-1}. Based on the rock magnetic properties, samples can be divided into two types. For Type 1, the coercivity (B_c) was very low (Figure 2(a)), and IRM was nearly saturated at an external field of <300 mT (Figure 2(c)), indicating soft magnetic minerals dominant. Their κ-T curves showed a distinct Curie temperature near 580°C, indicative of magnetite as the main magnetic carrier. The ratio of remanent coercivity to coercivity (B_{cr}/B_c) and the ratio of saturation remanent magnetization to saturation magnetization (M_{rs}/M_s) are 3.0–8.1 and 0.04–0.36 respectively, falling into the pseudo-single-domain and multi-domain range of

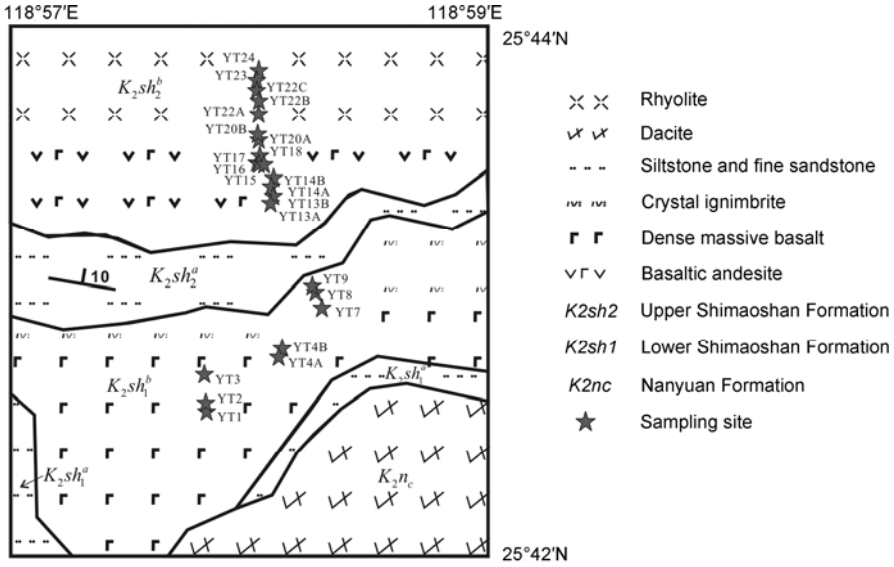


Figure 1 Simplified geological map for the sampling area. Stars stand for sampling sites. K_2Sh_2, Upper Shimaoshan Formation, K_2Sh_1, Lower Shimaoshan Formation, K_2n_c, Nanyuan Formation.

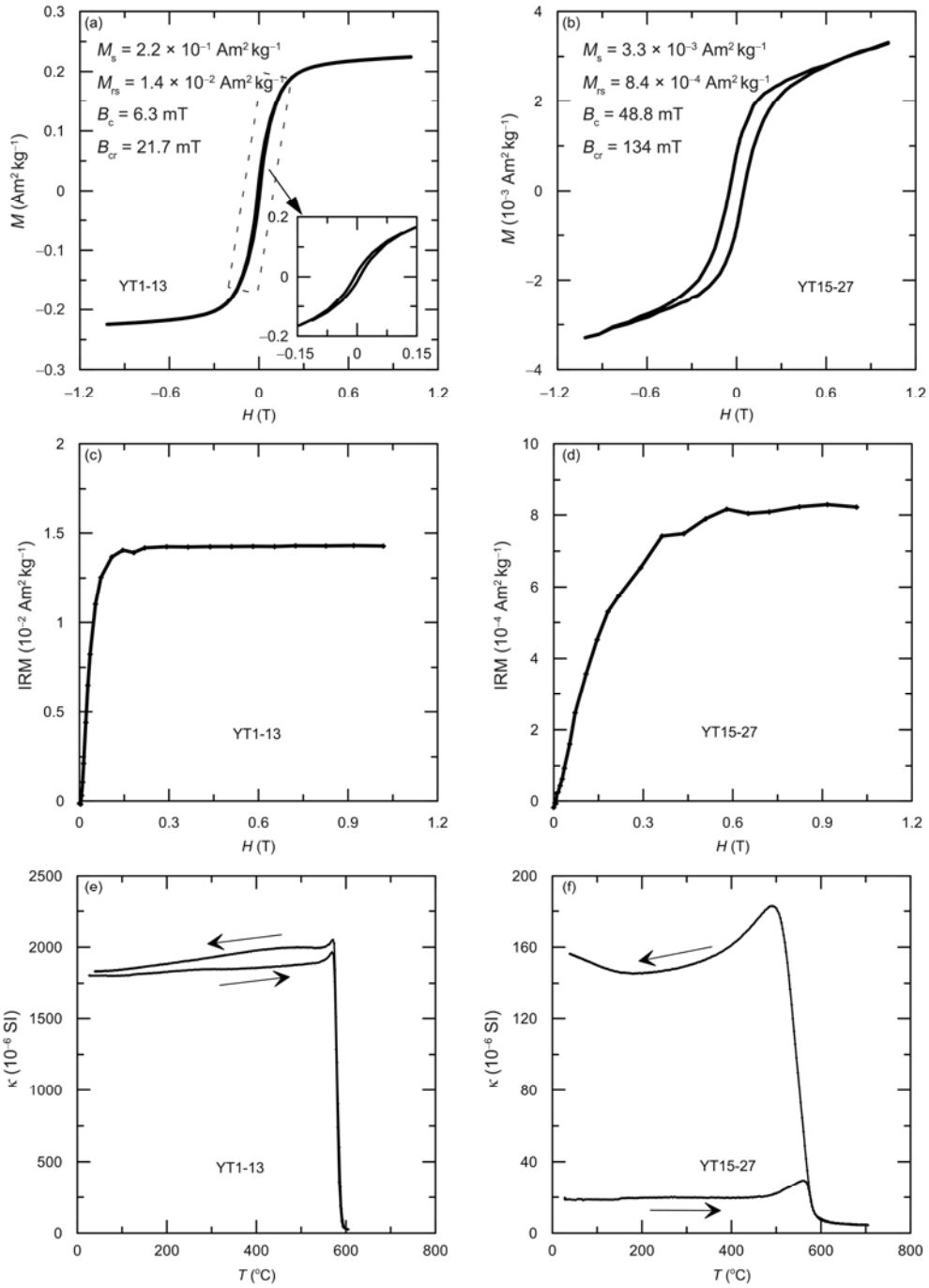


Figure 2 Room temperature hysteresis loops ((a), (b)), IRM acquisition curves ((c), (d)) and κ-T curves heated in argon ((e),(f)). M, magnetization; M_s, saturation magnetization; M_{rs}, saturation remanent magnetization; B_c, coercivity; B_{cr}, remanent coercivity; H, the external magnetic field.

the Day plot [28, 29]. For Type 2, the B_c was relatively high (Figure 2(b)), and IRM was not yet saturated even as the external field up to ~1 T (Figure 2(d)).Their κ-T curves showed a sharp drop near ~580°C, and continued decrease until ~680°C (Figure 2(f)), indicative of the mixture of magnetite and hematite as magnetic carriers

2.2 Paleomagnetism

Stepwise thermal demagnetization up to 585°C in 15–17 steps for Type 1 samples and up to 680°C in 19–21 steps for Type 2 samples at the interval of 50°C before 500°C and 10–25°C above 500°C was carried out using an ASC TD48 oven with a residual magnetic field less than 10 nT. Remanent measurements were made on a 2G cryogenic magnetometer situated in a field free space (<300 nT). As shown in Zijderveld plots (Figure 3), after removing the viscous remanent magnetization (VRM) at low temperatures, a single stable higher temperature magnetic component could be isolated for most samples, which were completely demag-

netized at ~580°C (magnetite) or ~680°C (hematite) in accordance with the rock magnetic results. The directions recorded by magnetite and hematite in one sample were coherent, implying that both of these two minerals were generated when the volcanic rocks formed. Some samples possessed two magnetic components. For example, the mean remanent direction of the middle-temperature (200–500°C) component (D_g/I_g = 3.9°/38.1°, α_{95} = 6.1°, N = 15) in site YT4B was significantly different from the high-temperature (> 500°C) component (D_g/I_g = 13.6°/53.9°, α_{95} = 2.5°, N = 17), likely due to the remagnetization. Therefore, the high-temperature component was preferred as the characteristic remanent magnetization (ChRM). The principal component direction was computed by a "least-squares fitting" technique [30], and at least five successive demagnetization steps trending towards the origin were selected to determine the ChRM. Specimens with a maximum angular deviation larger than 5° were also rejected for further analyses. Consequently, except YT23 with a success rate of 61%, probably caused by intensive alteration by weathering, the success rate of the other 22 sites exceeded 80%, and 383 samples had given reliable ChRM directions. Then averages for each site and statistical parameters were calculated assuming a Fisher distribution [31] (Table 1).

The directional data from all 23 sites recorded a normal polarity, in accordance with the section's age (87.9–88.6 Ma) within CNS. As shown in Figure 4(a), the average direction in geographic coordinates D_g/I_g = 3.0°/60.5° (α_{95} = 3.0°, N = 23) is significantly different from the present dipole field direction (D/I = 0°/43.9°) at the sampling locality, indicating that it was free of the recent remagnetization. To clarify whether these sites were paleomagnetically independent, we performed the Watson significant test [32] to compare the mean directions of adjacent individual sites. The results showed that YT13B and YT14A, YT16 and YT17, YT22C-YT24 were likely not paleomagnetically independent of each other. Consequently, there were 19 paleomagnectically independent sites in this sequence, resulting in the mean direction in stratigraphical coordinates (D_s/I_s = 4.5°/50.2°, α_{95} = 3.5°, N = 19) (Figure 4(b)). The corresponding paleomagnetic pole (83.1°N, 152.6°E, A_{95} = 3.9°, N = 19) for the section is at the 95% confidence level indistinguishable with the Eurasian reference pole [33] (82.2°N, 202.1°E, A_{95} = 5.2°) at 90 Ma (Figure 4(c)).

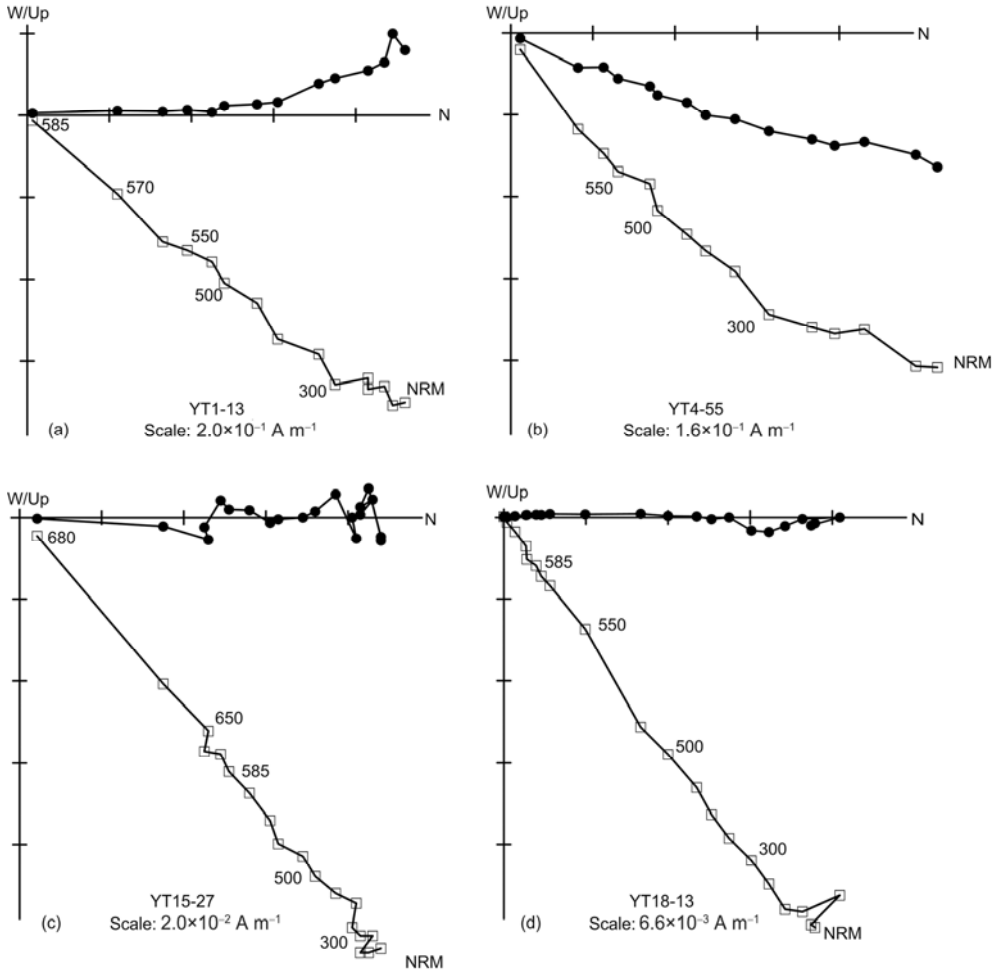


Figure 3 Representative orthogonal plots of progressive thermal demagnetization (in stratigraphic coordinates). Solid circles (open squares) are plotted on horizontal (vertical) planes. Thermal treatment levels are in degrees Celsius.

Table 1 Paleomagnetic data from the Yongtai section, Fujian Province[a)]

Location			*n*/*N*	D_g (°)	I_g (°)	D_s (°)	I_s (°)	α_{95}/A_{95}	*k*	Pole	
Site	Slon. (E)	Slat. (N)								Plon. (E)	Plat. (N)
YT1	118.9641°	25.7092°	17/18	5.6	46.1	6.1	36.1	2.6	184.8	253.0°	82.0°
YT2	118.9641°	25.7097°	23/23	12.1	50.1	11.6	40.1	2.8	114.4	221.7°	79.0°
YT3	118.9640°	25.7115°	12/12	13.8	64.1	12.5	54.1	4.0	120.7	166.4°	76.0°
YT4A	118.9694°	25.7132°	30/35	4.3	58.9	5.3	48.9	3.0	78.7	166.5°	83.8°
YT4B	118.9694°	25.7132°	17/17	13.6	53.9	12.8	44.0	2.5	209.2	205.8°	78.5°
YT7	118.9722°	25.7157°	7/7	10.0	58.3	9.8	48.3	2.3	720.0	184.3°	80.6°
YT8	118.9718°	25.7167°	8/8	13.7	61.3	12.6	51.3	2.4	525.4	176.5°	77.3°
YT9	118.9715°	25.7171°	17/17	9.6	65.5	9.4	55.5	1.2	855.6	154.7°	76.9°
YT13A	118.9687°	25.7233°	14/14	352.2	50.4	354.9	40.8	1.3	899.5	3.2°	84.8°
YT13B[a)]	118.9687°	25.7233°	18/18	0.5	56.8	2.2	46.9	1.8	350.7		
YT14A[a)]	118.9687°	25.7233°	13/13	2.3	57.8	3.7	47.8	2.0	429.3		
YT13B-14A			31/31	1.3	57.2	2.8	47.3	1.3	382.8	160.6°	86.3°
YT14B	118.9688°	25.7242°	31/37	1.1	61.8	3.0	51.8	2.4	115.4	139.6°	82.8°
YT15	118.9682°	25.7247°	23/27	21.1	58.5	18.6	48.6	5.2	34.7	191.6°	73.1°
YT16[b)]	118.9678°	25.7247°	26/27	355.1	60.0	358.2	50.2	2.9	95.4		
YT17[b)]	118.9678°	25.7251°	14/15	351.9	58.8	355.6	49.2	2.6	233.2		
YT16-17			40/42	354.0	59.6	357.3	49.9	2.1	120.7	94.2°	84.5°
YT18	118.9679°	25.7253°	19/21	356.2	63.8	359.4	54.0	1.5	518.7	115.8°	81.1°
YT20A	118.9678°	25.7263°	11/12	4.5	74.0	6.2	64.0	9.7	23.1	131.4°	69.4°
YT20B	118.9678°	25.7263°	17/20	337.5	62.5	345.0	53.7	3.5	103.4	65.9°	74.5°
YT22A	118.9679°	25.7278°	19/20	348.3	63.2	353.3	53.7	1.7	414.4	86.4°	79.7°
YT22B	118.9678°	25.7286°	7/7	350.6	59.9	354.7	50.2	2.7	502.5	78.7°	83.0°
YT22C[c)]	118.9677°	25.7294°	13/14	5.8	65.7	6.7	55.7	4.2	100.1		
YT23[c)]	118.9673°	25.7299°	14/23	7.6	66.1	8.0	56.1	4.2	91.9		
YT24[c)]	118.9661°	25.7306°	13/13	10.1	66.3	9.8	56.3	3.3	161.3		
YT22C-23-24			40/50	7.8	66.0	8.1	56.0	2.1	115.0	149.6°	77.1°
Paleomagnetic pole								3.9	76.5	152.6°	83.1°

a) Slat. (Plat.) and Slon. (Plon): the latitude and longitude of the site (Pole), D_g (D_s), I_g (I_s): the declination and inclination in geographical (stratigraphical) coordinates, *n*/*N*: the number of samples used to calculate the mean and measured. α_{95} (A_{95}): the radius of cone at 95% confidence level about the mean direction (pole), *k:* the best estimate of the precision parameter. a), b) and c) stand for paleomagnetically correlated sites, respectively. The strata's dip and its corresponding direction for Yongtai section are 10° and 9°.

3 Discussion

3.1 Paleomagnetic secular variations

Geocentric axial dipole is one of the primary hypotheses for paleomagnetism. It assumes the geomagnetic pole completely overlaps the geographical pole. However, the instantaneous geomagnetic pole in fact temporarily deviates from the geographical pole, and merely statistically coincides with it after averaging out the geomagnetic secular variation on time scales of 10^4 and 10^5 years [34]. If a set of paleomagnetic data is free of paleomagnetic secular variation PSV influence, their corresponding virtual geomagnetic poles (VGP) will bear some kind of statistical characteristics such as Fisher distribution [35, 36] and the latitudinal dependence of angular dispersion [37].

To test VGPs Fisher distribution, the quantile-quantile (Q-Q) method [38] is used. In general, it can test whether VGPs' longitudes and colatitudes as viewed about the mean VGPs are uniformly distributed and exponentially distributed, respectively. We firstly transposed the 19 VGPs longitudes and colatitudes at Yongtai section to make the mean (paleomagnetic pole) located at the North Pole. Secondly, the VGPs' longitudes and colatitudes quantiles were calculated and compared with the theoretical uniform and exponential quantiles. As shown in Figure 5, the corresponding values nearly fell on a straight line, indicating that VGPs fairly followed these two distributions [38]. Furthermore, it shows that the statistical parameters M_u (= 0.7926) for uniform distribution and M_e (= 0.4210) for exponential distribution are less than the 95% confidence value 1.207 for uniform distribution and 1.094 for exponential distribution [38], respectively. It suggests that these VGPs obeyed the Fisher distribution, indicating that the paleomagnetic results of this study sufficiently averaged out possible PSV

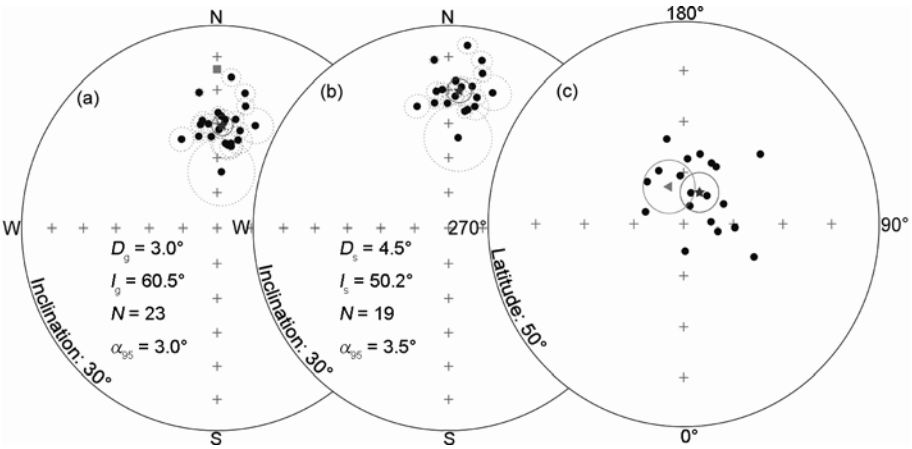


Figure 4 Equal-area projections of site-mean ChRM with 95% confidence circle for Yongtai section in geographic (a) and stratigraphic (b) coordinates. Solid circles represent directions plotted onto the lower hemisphere. Star denotes the overall mean direction, and square in (a) denotes the present dipole field direction at the sampling locality. (c) Equal-area projection of virtual geomagnetic poles (solid circles) and their and its corresponding paleomagnetic pole (star) for Yongtai section. Triangle represents the Eurasian reference pole at 90 Ma.

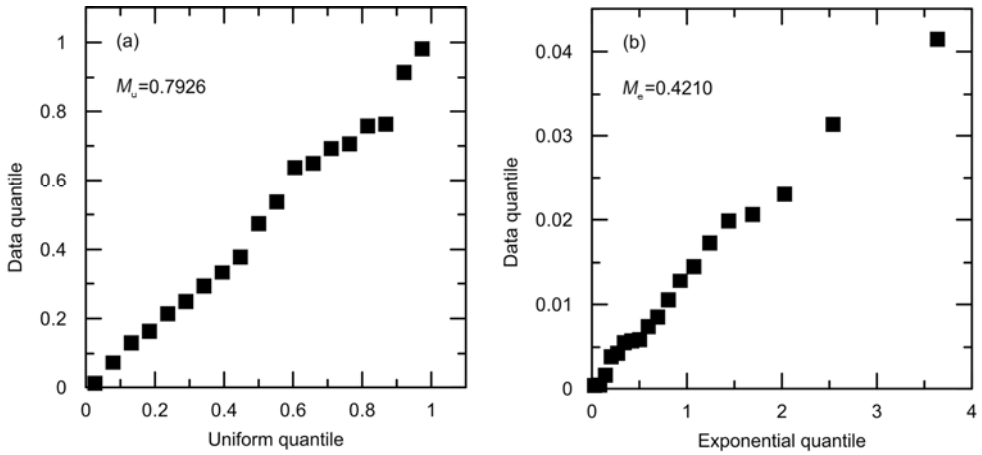


Figure 5 Quantile-quantile plot of the longitude (a) and the colatitude (b) of VGP when viewed about the mean VGPs at Yongtai section.

influence.

The scatter S_B is another important proxy for assessing PSV. Based on the observations of the present geomagnetic field (IGRF65), McFadden et al. [37] put forward a Model G to describe the scatter S_B as a function of paleolatitude (λ) using two shape parameters a and b:

$$S_B^2 = (b\lambda)^2 + a^2 , \quad (1)$$

where a and b represented variations in the equatorially symmetric and asymmetric spherical harmonic terms, respectively. Relied on the paleomagnetic database, McFadden et al. [39] calculated the theoretical parameters a and b for 0–5 Ma and other geological periods corresponding to different geomagnetic reversal frequency during 0–195 Ma. Therefore, for a set of paleomagnetic data, a comparison of the S_B with the corresponding theoretical value can be used to evaluate whether the data were affected by PSV.

In practice, the scatter S_B is obtained by removing the within-site dispersion S_{wi} from the angular dispersion between VGPs and their mean or the geographic pole. Usually, VGPs are from rock units of similar age and the same geographic region and the scatter S_B from N units is described as

$$S_B = \left[\frac{1}{N-1} \sum_{i=1}^{N} (\Delta_i^2 - \frac{S_{W_i}^2}{n_i}) \right]^{1/2} , \quad (2)$$

where Δ_i gives the angular distance of the ith VGP from their mean or the geographic pole. In order to exclude the influence of excursional data, Δ_i is confined to be less than the cutoff value 45°. S_{W_i} is often correlated with the ith VGP and likely caused by random errors associated with sampling and measuring process. Thus, to calculate S_{W_i}, the precision parameter k of direction space and the paleolatitude λ of sampling site are estimated first, and then S_{W_i}

is approximately calculated by [40]:

$$S_{W_i} = \frac{81}{\sqrt{k\left[\frac{1}{8}(5+18\sin^2\lambda+9\sin^4\lambda)\right]^{-1}}}. \quad (3)$$

The corresponding paleolatitude for the Yongtai section was 31.4°N, and its S_B was 9.0 with the 95% confidence level (7.2, 10.8). The result was indistinguishable from the expected value [41] in CNS at the 95% confidence level for this paleolatitude (Figure 6), indicative of paleomagnetic results free of PSV influence.

The PSV can be used to assess the stability of the geomagnetic field [41–43]. Biggin et al. [41] put forward a set of criteria to select global paleomagnetic database for PSV analysis: (1) data were from lavas or welded tuff, (2) the samples should have undergone stepwise demagnetization and isolated the ChRM, and (3) the amount of paleomagnetically individual rock units was ⩾7. Then they analyzed the PSV data in the CNS and deduced the param- eters a = 9.9 and b = 0.26 for describing the shape of VGPs' scatter curves during that period. Compared with the shape parameters (a = 14.6, b = 0.20) in the period 0–5 Ma (when average reversal frequency was ~4.0 Ma^{-1}), a was lower and b was higher in CNS, indicating that the earth's field is more asymmetric in the stable polarity period, which is consistent with the result of geodynamo simulation [44].

Paleomagnetic data of this study and an earlier study by Zhu et al. [45] on the volcanic sequences at Suhongtu section in Inner Mongolia, China (Ar/Ar age: 110–114 Ma) provide new data from the mid-latitude area. To keepconsistent with Biggin et al.'s analyses [41], the above criteria were used to select data. Consequently, S_B for Suhongtu section was 10.0 with the 95% confidence level (8.5, 11.4), with the corresponding paleolatitude λ of 39.3°N. Then, the two new data were plotted on the correlation diagram between S_B and λ. Based on the data with $\lambda < 40°$ and $\lambda > 40°$, we fit the curve and obtained the shape parameters a = 10.7, b = 0.09 for lower latitudes and a = 7.7, b = 0.28 for higher latitudes, respectively. It suggests that S_B was contributed mainly by the symmetric terms at lower latitudes whereas both the symmetric and asymmetric terms at higher latitudes contributed to S_B. The discrepancy of S_B between lower and higher latitudes might arise from unknown earth's interior dynamo process or the limited amount of paleomagnetic data.

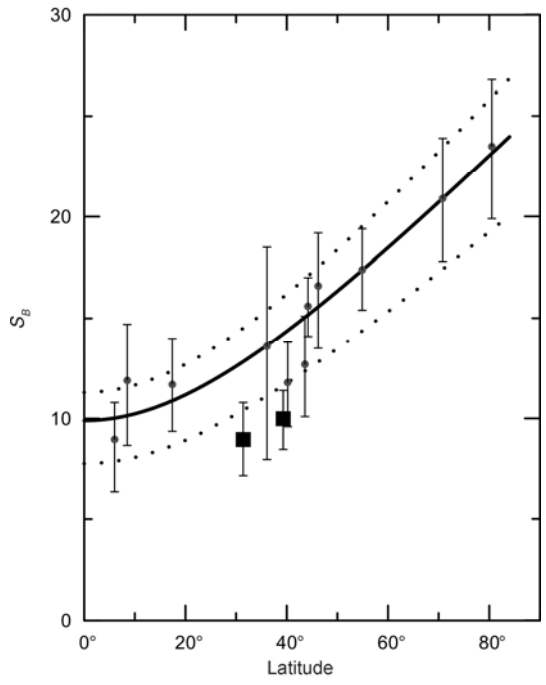


Figure 6 The relationship between the scatter S_B and the latitude during the Cretaceous Normal Superchron. The solid line and dotted lines represent the fitting curve and its 95% confidence level. Filled squares stand for the Yongtai section and Suhongtu section (see text).

3.2 Paleolatitude of the South China Block

The consistence between the new paleomagnetic pole from Yongtai section and the Eurasian reference pole of 90 Ma suggested that the sampling area did not experience an obvious southward movement relative to the Eurasian plate during the Late Cretaceous. Furthermore, we compiled available paleomagnetic data from the Hong Kong's dyke at ~80 Ma (69.3°N, 211.2°E, A_{95} = 8.9°, N = 4) [46] and eastern Zhejiang Province's volcanic rocks between 109–118 Ma (76.3°N, 194.5°E, A95 = 7.6°, N = 15) [27] to calculate the paleolatitudes of SCB using Yongtai section (25.7°N, 119°E) as a reference site. The deduced paleolatitudes between 80 and 118 Ma showed no significant distinction from the expected paleolatitude calculated from the Eurasian apparent polar wander path (APWP) [33] at the 95% confidence level (Figure 7), indicating that no obvious southward movement of SCB occurred during the Cretaceous.

To probe the discrepancy between red beds, we reexamined the red-beds paleomagnetic results in SCB with more strict criteria: (1) In order to avoid the influence of tectonic movement, the sampling areas undergoing the local deformation were excluded. For example, in the western part of SCB, only the combined paleomagnetic pole (72.8°N, 241.1°E, A_{95} = 6.3, N = 16) from Guanyin section (N = 7) and Feixianguan section (N = 9) in Sichuan Province was retained, where the pole from the Feixianguan section was corrected by 8° of local rotations [12, 15]. (2) Only the data from Late Cretaceous were discussed. (3) The amount of sampling sites must be greater than 5. The data satisfying the criteria above are listed in Table 2.

Based on the paleomagnetic poles from red beds, the mean paleomagnetic poles for the middle-western and eastern part of SCB in the Late Cretaceous (the pole from Mayuan basin was excluded in calculation) were 73.0°N, 239.7°E (A_{95} = 4.2, N = 4) and 79.9°N, 189.7°E (A_{95} = 4.7, N = 4), respectively. Their corresponding declination and inclination to the sampling area (25.7°N, 119°E) were

Table 2 Late Cretaceous poles from South China Block[a)]

Area	Location		N	Pole		A_{95}	λ (N)	Reference
	Slon. (E)	Slat. (N)		Plon. (°E)	Plat. (°N)			
Zhejiang	119.9°	28.4°	19	214.3	81	6.4	24.5°	[13]
Jiangsu	119°	32°	10	172.6	76.3	10.3	33.2°	[47]
Fujian	117.8°	26.4°	22	201.7	79.7	5.8	26.6°	[13]
Guangdong	115.8°	24.1°	9	177.7	80.8	8	30.2°	[13]
Hunan	109.9°	27.9°	17	168.1	83.5	4	29.8°	[14]
Jiangxi	114.9°	25.9°	14	225.1	74.4	5.2	20.5°	[11]
Hunan	112.6°	26.9°	26	236.3	71.9	4.7	16.5°	[10]
Hubei	111.7°	30.7°	18	254.1	71.7	5.6	12.2°	[16]
Sichuan	102.9°	30°	16	241.1	72.8	6.3	15.8°	[15]
Fujian	119°	25.7°	19	152.6	83.1	3.9	31.4°	This study
Eurasian reference pole (70–100 Ma)				198.5	81.5	2.2	26.9°	[33]

a) Slat. (Plat.) and Slon. (Plon.): The latitude and longitude of the site (pole), N: the number of sampling sites. λ is the expected paleolatitude calculated from the common site (25.7°N, 119°E). Except for this study, all the other data are from red beds.

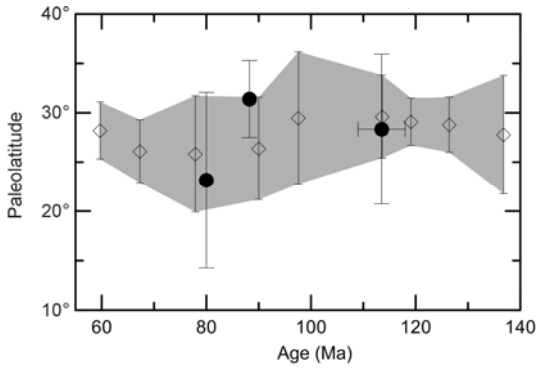


Figure 7 Paleolatitudes calculated from the poles for volcanic rocks in the South China Block (circles) and the reference Eurasian APWP (diamonds) for a common site (25.7°N, 119°E) in the Cretaceous.

15.2°±4.4°, 30.3°±6.8° for the middle-western part of SCB and 10.9°±5.4°, 47.5°±5.6° for the eastern part of SCB. It is clear that the declinations of the two parts of SCB are consistent with each other at the 95% confidence level, suggesting that no obvious rotation occurred between the middle-western and eastern part of SCB in the Late Cretaceous. However, their inclinations are significantly different, and their corresponding paleolatitudes for the common site (25.7°N, 119°E) vary between 12.2°N and 33.2°N. Three out of five are lower than the expected from the Eurasian reference pole in the Late Cretaceous (70–100 Ma) [33], which might have resulted from the flattening of the red-beds inclination. By analyzing the anisotropy of IRM for red-beds samples Wang and Yang [11] recently found that samples in Jishui, Jiangxi Province suffered from inclination flattening.

4 Conclusions

Paleomagnetic data from 19 sampling sites of a Late Cretaceous volcanic sequence in Yongtai County, Fujian Province were obtained. The Fisher distribution of VGPs and the consistence of S_B with the expected value of the PSV curve in the Cretaceous Normal Superchron indicate that the yielded paleomagnetic pole is free of PSV influence. The new paleomagnetic pole at 83.1°N, 152.6°E (N = 19, A_{95} = 3.9°), in well accordance with that of the Eurasian APWP, suggests no obvious southward movement relative to the Eurasian plate during the Cretaceous.

Paleomagnetic secular variation analyses suggest that the symmetric terms played an important role in S_B at lower latitudes and both the symmetric and asymmetric terms contributed to it at higher latitudes during the Cretaceous Normal Superchron.

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